

# Mathematics II

A complete student textbook for determinants, matrices, vector algebra, integral calculus, applications of integration and first-order differential equations—with methods, diagrams, solved problems, practice, quick revision and full model answers.

Course Code

**2002**

Programme

**Diploma in Engineering and Technology**

Semester

**2**

Credits

**4**

Category

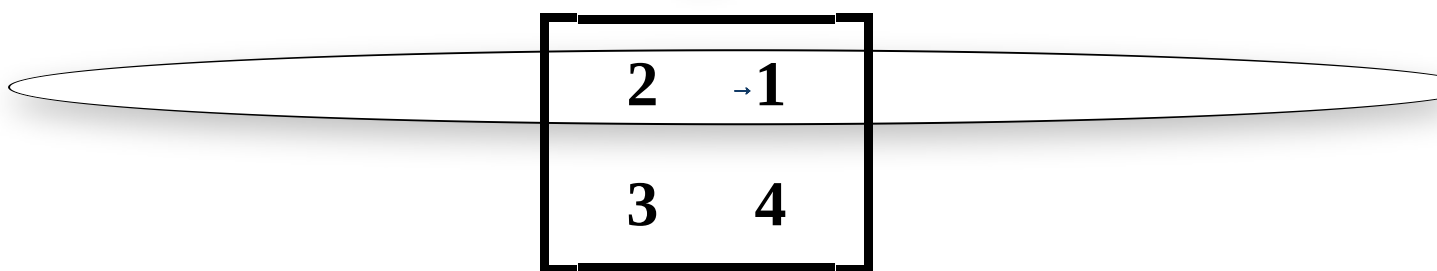
**Basic Science**

Periods

**4/week • 60/semester**

## Student promise

Every official syllabus item is developed through concepts, methods, worked examples and examination practice. The handbook does not add advanced topics beyond the level needed to understand and apply the prescribed syllabus.



## Four-module learning path

Determinants, matrices, Cramer's rule and inverse method

Polytechnic Study Hub

Vectors, dot product, cross product, work and moment

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## COURSE OVERVIEW

# Mathematics as an engineering problem-solving language

Mathematics II continues the algebra and calculus foundation developed in Mathematics I. It teaches four connected tools: matrices for simultaneous systems, vectors for magnitude and direction, integration for accumulation, and differential equations for changing systems.

## Linear systems

Determinants and matrices organise several equations and provide systematic methods such as Cramer's rule and inverse matrices.

## Magnitude and direction

Vectors describe force, displacement, velocity and moment. Dot and cross products translate geometry into calculation.

## Accumulation and change

Integration finds accumulated quantity and area. Differential equations describe relations involving a quantity and its rate of change.

## Where diploma students use these ideas

### Electrical networks

Simultaneous equations for currents and voltages.

### Mechanics

Force resolution, work and turning moment.

### Geometry and design

Area enclosed by curves and coordinate axes.

## Dynamic systems

Simple growth, decay and linear first-order models.

### HOW TO USE THIS HANDBOOK

# Understand the method before memorising the formula

## 1. Read

Begin with the meaning, notation and conditions of the concept.

## 2. Follow

Trace each worked example line by line, especially signs and limits.

## 3. Practise

Close the solution and repeat the problem independently.

## 4. Check

Use substitution, differentiation, dimensions or geometric sense to verify.

### Examination method

Write the formula or method first, show essential steps, simplify carefully, box the final answer, and include units for work, moment and engineering quantities.

### TABLE OF CONTENTS

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Formula Bank



Method Bank



Diagram Bank



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Expected Questions



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Quick Revision



Model Question Paper



Full Model Answers



Answer Key



References



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#### OFFICIAL SYLLABUS ANALYSIS

## Course objective and prerequisite

### Official objective

To give a comprehensive coverage at an introductory level to Determinants and Matrices, Integral Calculus, basic elements of Vector Algebra and First Order Differential Equations as applicable to solve engineering problems.

### Student-friendly meaning

The course is not about memorising isolated formulas. It develops four practical methods that help organise equations, represent directed quantities, calculate accumulation and model change.

### Prerequisite

**Course:** Mathematics I, Semester 1

**Required topics:** Complex Numbers, Coordinate Geometry, Trigonometry, Limits and Derivatives.

### Derivatives

needed to understand integration as the reverse operation.

### Coordinate geometry

needed for vectors, position vectors and areas.

### Trigonometry

needed for vector angles and standard integrals.

### Algebra

needed for determinants, matrices and differential equations.

## OFFICIAL COURSE OUTCOMES

# What students must be able to do

| CO  | Official outcome                                                                                 | Duration | Cognitive level | Student meaning                                                   |
|-----|--------------------------------------------------------------------------------------------------|----------|-----------------|-------------------------------------------------------------------|
| CO1 | Make use of Determinants and Matrices in finding the solutions of a linear system.               | 12 h     | Applying        | Evaluate determinants and solve simultaneous equations.           |
| CO2 | Identify the concept of scalar and vector quantities and apply it in engineering problems.       | 16 h     | Applying        | Use vector operations for work, angle and moment.                 |
| CO3 | Build the concept of integration as the inverse operation of differentiation.                    | 18 h     | Applying        | Integrate using standard results, substitution, parts and limits. |
| CO4 | Apply integration techniques to solve different engineering problems and differential equations. | 12 h     | Applying        | Find area and solve simple first-order differential equations.    |
| —   | Series tests                                                                                     | 2 h      | —               | One hour after Module 2 and one hour after Module 4.              |

## CO-PO mapping

| Course outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | Interpretation         |
|----------------|-----|-----|-----|-----|-----|-----|-----|------------------------|
| CO1            | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1 |
| CO2            | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1 |
| CO3            | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1 |
| CO4            | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1 |

### Mapping scale

3 - Strongly mapped; 2 - Moderately mapped; 1 - Weakly mapped.

## OFFICIAL MODULE ROADMAP

# Module outcomes, hours and level

| Module outcome | Description                                                             | Hours | Level         |
|----------------|-------------------------------------------------------------------------|-------|---------------|
| M1.01          | Outline the second and third order determinants.                        | 5     | Understanding |
| M1.02          | Apply Cramer's Rule to solve a linear system of equations.              | 2     | Applying      |
| M1.03          | Solve a system of equations in two unknowns by inverse matrix.          | 5     | Applying      |
| M2.01          | Compare vector and scalar quantities.                                   | 4     | Understanding |
| M2.02          | Explain vector operations, scalar product and vector product.           | 6     | Understanding |
| M2.03          | Apply dot and cross products to work and moment.                        | 6     | Applying      |
| Series Test I  | Modules 1 and 2                                                         | 1     | —             |
| M3.01          | Relate integration to reverse differentiation and rules of integration. | 3     | Understanding |
| M3.02          | Solve integration problems by substitution and by parts.                | 8     | Applying      |
| M3.03          | Explain definite integrals and solve problems.                          | 7     | Applying      |
| M4.01          | Apply integration to find area bounded by a curve and the axes.         | 7     | Applying      |
| M4.02          | Outline and solve first-order, first-degree differential equations.     | 5     | Applying      |
| Series Test II | Modules 3 and 4                                                         | 1     | —             |

## MODULE 1 • CO1 • 12 HOURS

# Determinants and Matrices

Second- and third-order determinants, matrix operations, Cramer's rule for three variables, inverse of a matrix, and inverse method for two variables.

## 1. Determinants: meaning, order and notation

A determinant is a single number associated with a square arrangement of numbers. It helps test whether a linear system has a unique solution and is used in inverse matrices and Cramer's rule.

Second order:  $|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

**Malayalam Support:** Determinant ഒരു square array-യിൽ നിന്ന് ലഭിക്കുന്ന ഒരു സംഖ്യയാണ്. Equations-ന് unique solution ഉണ്ടോ എന്ന് പരിശോധിക്കാനും inverse കണ്ടെത്താനും ഇത് ഉപയോഗിക്കുന്നു.

### Order

A determinant with n rows and n columns has order n. A 2×2 determinant is second order; a 3×3 determinant is third order.

## Rows and columns

Horizontal entries form rows; vertical entries form columns. Determinants are defined only for square arrays.

### Solved Example 1: Second-order determinant

Evaluate  $\begin{vmatrix} 4 & 3 \\ 2 & 5 \end{vmatrix}$ .

$$|A| = (4 \times 5) - (3 \times 2) = 20 - 6 = 14$$

**Answer:** 14.

## 2. Third-order determinants

A third-order determinant may be expanded along any row or column. Signs of cofactors follow the checkerboard pattern:

$$\text{Sign pattern} = \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$\text{For } A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, |A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$

#### Common mistake

The middle term is negative. Write the sign pattern before expanding.

|                                                                                                                                                                                                                                                                                                                                                                                                                |   |                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-around; font-size: 24px;"> <span>a</span><span>b</span><span>c</span> </div> <div style="display: flex; justify-content: space-around; font-size: 24px; margin-top: 10px;"> <span>d</span><span>e</span><span>f</span> </div> <div style="display: flex; justify-content: space-around; font-size: 24px;"> <span>g</span><span>h</span><span>i</span> </div> | → | <div style="margin-bottom: 10px;"><b><math>a(ei - fh)</math></b></div> <div style="margin-bottom: 10px;"><b><math>- b(di - fg)</math></b></div> <div><b><math>+ c(dh - eg)</math></b></div> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Expansion along the first row. Each entry multiplies the determinant left after deleting its row and column.

### Solved Example 2: Third-order determinant

Evaluate  $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 0 & 1 \end{bmatrix}$ .

$$\Delta = 1(1 \times 1 - 4 \times 0) - 2(0 \times 1 - 4 \times 2) + 3(0 \times 0 - 1 \times 2)$$

$$\Delta = 1 - 2(-8) + 3(-2) = 1 + 16 - 6 = 11$$

**Answer:** 11.

## 3. Matrices

A matrix is a rectangular arrangement of numbers in rows and columns. Unlike a determinant, a matrix itself is an array, not a single numerical value.

$A = [a_{ij}]_{m \times n}$  means m rows and n columns.

| Type                   | Definition                                 | Example                      |
|------------------------|--------------------------------------------|------------------------------|
| Row matrix             | One row                                    | [2 5 -1]                     |
| Column matrix          | One column                                 | [2; 5; -1]                   |
| Rectangular            | Rows $\neq$ columns                        | 2 $\times$ 3 matrix          |
| Square                 | Rows = columns                             | 2 $\times$ 2 or 3 $\times$ 3 |
| Zero/null              | Every entry is zero                        | O                            |
| Diagonal               | Non-diagonal entries are zero              | diag(2,3)                    |
| Scalar                 | Diagonal entries equal                     | diag(4,4)                    |
| Identity/unit          | Diagonal entries 1; others 0               | I                            |
| Upper/lower triangular | Entries below/above main diagonal are zero | Square triangular matrix     |

## 4. Equality, addition and subtraction

Two matrices are equal only when they have the same order and every corresponding entry is equal. Addition and subtraction are possible only for matrices of the same order.

$$[a_{ij}] \pm [b_{ij}] = [a_{ij} \pm b_{ij}]$$

### Solved Example 3: Matrix addition

Let  $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & 5 \\ -2 & 0 \end{bmatrix}$ .

$$A+B = \begin{bmatrix} 3 & 4 \\ 1 & 4 \end{bmatrix}; A-B = \begin{bmatrix} 1 & -6 \\ 5 & 4 \end{bmatrix}$$

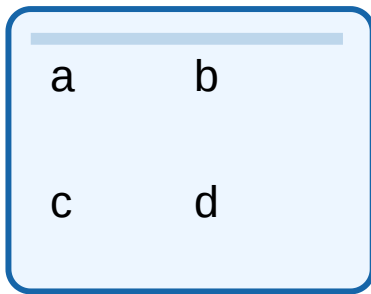


## 5. Matrix multiplication

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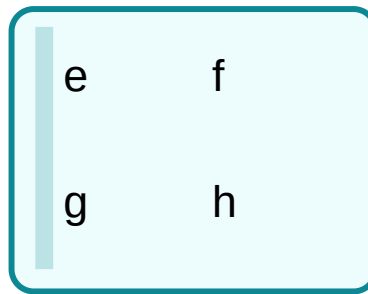
The product  $AB$  exists when the number of columns of  $A$  equals the number of rows of  $B$ . If  $A$  is  $m \times n$  and  $B$  is  $n \times p$ , then  $AB$  is  $m \times p$ .

$$(AB)_{ij} = \text{sum of products of row } i \text{ of } A \text{ with column } j \text{ of } B.$$



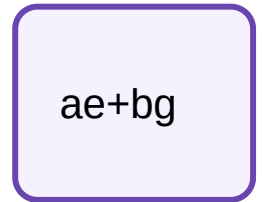
Row of A

$\times$



Column of B

$=$



Entry (1,1)

Row-by-column rule. Matrix multiplication is generally not commutative:  $AB$  may not equal  $BA$ .

### Solved Example 4: Matrix product

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}.$$

$$AB = \begin{bmatrix} 1 \times 2 + 2 \times 1 & 1 \times 0 + 2 \times 5 \\ 3 \times 2 + 4 \times 1 & 3 \times 0 + 4 \times 5 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}$$

## 6. Inverse of a $2 \times 2$ matrix

A square matrix  $A$  has an inverse only when  $|A| \neq 0$ . Such a matrix is nonsingular.

$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } A^{-1} = 1/(ad-bc) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

### Solved Example 5: Inverse

$$\text{Find the inverse of } A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}.$$

$$|A| = 3 \times 1 - 1 \times 2 = 1.$$

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$\text{Check: } AA^{-1} = I.$$

## 7. Solving two equations by inverse matrix

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$$AX=B \Rightarrow X=A^{-1}B$$

### Solved Example 6: Inverse method

Solve  $3x+y=7$  and  $2x+y=5$ .

$$A=\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}, X=\begin{bmatrix} x \\ y \end{bmatrix}, B=\begin{bmatrix} 7 \\ 5 \end{bmatrix}$$

$$X=A^{-1}B=\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 7 \\ 5 \end{bmatrix}=\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

**Answer:**  $x=2, y=1$ .

## 8. Cramer's rule for three variables

For a system  $AX=B$  with  $\Delta=|A|\neq 0$ , replace one coefficient column at a time by the constants column.

$$x=\Delta_x/\Delta, y=\Delta_y/\Delta, z=\Delta_z/\Delta$$

- 1 Write coefficients in the same variable order.
- 2 Evaluate coefficient determinant  $\Delta$ .
- 3 Form  $\Delta_x$ ,  $\Delta_y$  and  $\Delta_z$  by replacing the corresponding column.
- 4 Divide each replaced determinant by  $\Delta$ .
- 5 Substitute values into the original equations to verify.

### Solved Example 7: Cramer's rule

Solve  $x+y+z=6$ ,  $2x-y+z=3$ ,  $x+2y-z=2$ .

Coefficient determinant:

$$\Delta=\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{bmatrix}=7$$

$$\Delta_x=7, \Delta_y=14, \Delta_z=21$$

**Answer:**  $x=1, y=2, z=3$ .

## Module 1 exam tip

For Cramer's rule, preserve the variable order x, y, z in every determinant. For inverse method, first confirm  $|A| \neq 0$ .

### Module 1 Quick Check and Answers

1. Evaluate  $|5 \ 2; \ 3 \ 1|$ . **Answer:** -1.
2. Can a  $2 \times 3$  matrix have a determinant? **Answer:** No.
3. State the order of AB if A is  $2 \times 3$  and B is  $3 \times 4$ . **Answer:**  $2 \times 4$ .
4. When does  $A^{-1}$  exist? **Answer:** When A is square and  $|A| \neq 0$ .

## MODULE 2 • CO2 • 16 HOURS • SERIES TEST I

# Vector Algebra

Scalars and vectors, types of vectors, algebra, position vector, scalar product, vector product, work and moment.

## 1. Scalars and vectors

### Scalar

A quantity having magnitude only. Examples: mass, time, temperature, energy and speed.

### Vector

A quantity having magnitude and direction and obeying vector addition. Examples: displacement, velocity, acceleration, force and moment.

**Malayalam Support:** Scalar-ന് magnitude മാത്രം. Vector-ന് magnitude-ഉം direction-ഉം ഉണ്ട്. Force, displacement എന്നിവ vectors ആണ്.

## 2. Vector notation, magnitude and unit vector

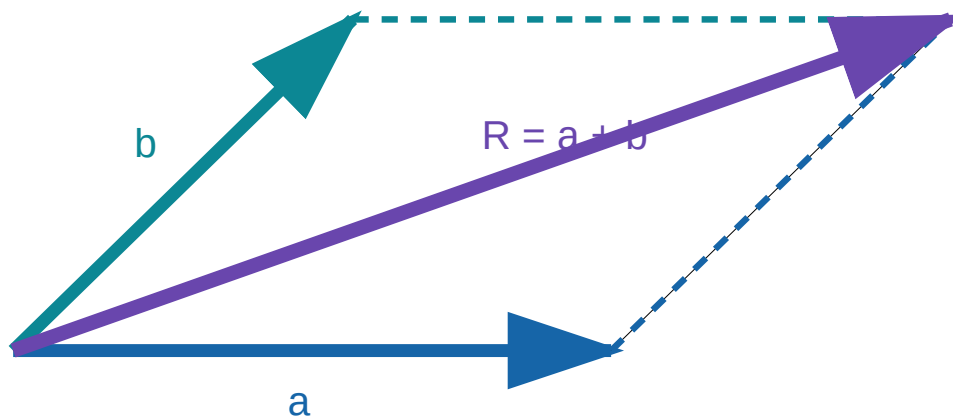
$$\mathbf{a} = a_x \mathbf{i} + a_y \mathbf{j} + a_z \mathbf{k}$$

$$|\mathbf{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}, \quad \hat{\mathbf{a}} = \mathbf{a}/|\mathbf{a}|$$

| Type                | Meaning                                        |
|---------------------|------------------------------------------------|
| Zero vector         | Magnitude zero; direction not definite.        |
| Unit vector         | Magnitude one.                                 |
| Equal vectors       | Same magnitude and same direction.             |
| Negative vectors    | Same magnitude, opposite directions.           |
| Parallel/collinear  | One is a scalar multiple of the other.         |
| Like/unlike vectors | Parallel vectors with same/opposite direction. |
| Position vector     | Vector from origin to a point.                 |

### 3. Vector addition and subtraction

$$\mathbf{a} + \mathbf{b} = (a_x + b_x)\mathbf{i} + (a_y + b_y)\mathbf{j} + (a_z + b_z)\mathbf{k}$$



Parallelogram law: the diagonal from the common tail represents the resultant  $\mathbf{a} + \mathbf{b}$ .

#### Solved Example 8: Magnitude and unit vector

For  $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}$ ,  $|\mathbf{a}| = \sqrt{9 + 16 + 144} = 13$ .

$$\hat{\mathbf{a}} = (3/13)\mathbf{i} - (4/13)\mathbf{j} + (12/13)\mathbf{k}$$

### 4. Position vector

If  $P(x, y, z)$ ,  $\mathbf{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ .

$$\text{Vector } \mathbf{AB} = \mathbf{OB} - \mathbf{OA} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$$

#### Solved Example 9: Vector between points

$A(1, 2, -1)$ ,  $B(4, -2, 3)$ .

$$\mathbf{AB} = (4-1)\mathbf{i} + (-2-2)\mathbf{j} + (3+1)\mathbf{k} = 3\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$$

$$|\mathbf{AB}| = \sqrt{(9+16+16)} = \sqrt{41}.$$

## 5. Scalar or dot product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta = a_x b_x + a_y b_y + a_z b_z$$

### Angle

$$\cos\theta = (\mathbf{a} \cdot \mathbf{b}) / (|\mathbf{a}||\mathbf{b}|)$$

### Perpendicular test

For nonzero vectors,  $\mathbf{a} \cdot \mathbf{b} = 0$  means  $\mathbf{a}$  is perpendicular to  $\mathbf{b}$ .

### Solved Example 10: Dot product and angle

$\mathbf{a} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ ,  $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ .

$$\mathbf{a} \cdot \mathbf{b} = 2 + 2 - 4 = 0.$$

**Conclusion:** The vectors are perpendicular.

## 6. Work done by a constant force

$$\text{Work } W = \mathbf{F} \cdot \mathbf{s} = |\mathbf{F}||\mathbf{s}|\cos\theta$$

Only the component of force along the displacement contributes to work.

### Solved Example 11: Work

A force  $\mathbf{F} = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$  N moves a body through  $\mathbf{s} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$  m.

$$W = \mathbf{F} \cdot \mathbf{s} = 4 \times 2 + (-2) \times 1 + 3 \times (-4) = 8 - 2 - 12 = -6 \text{ J}$$

**Answer:** -6 J. Negative work means the force has a component opposing displacement.

## 7. Vector or cross product

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin\theta \hat{n}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

### Direction

The direction is perpendicular to both  $a$  and  $b$  and follows the right-hand rule. Also  $a \times b = -(b \times a)$ .

## Solved Example 12: Cross product

$a = i + 2j + 3k$ ,  $b = 2i - j + k$ .

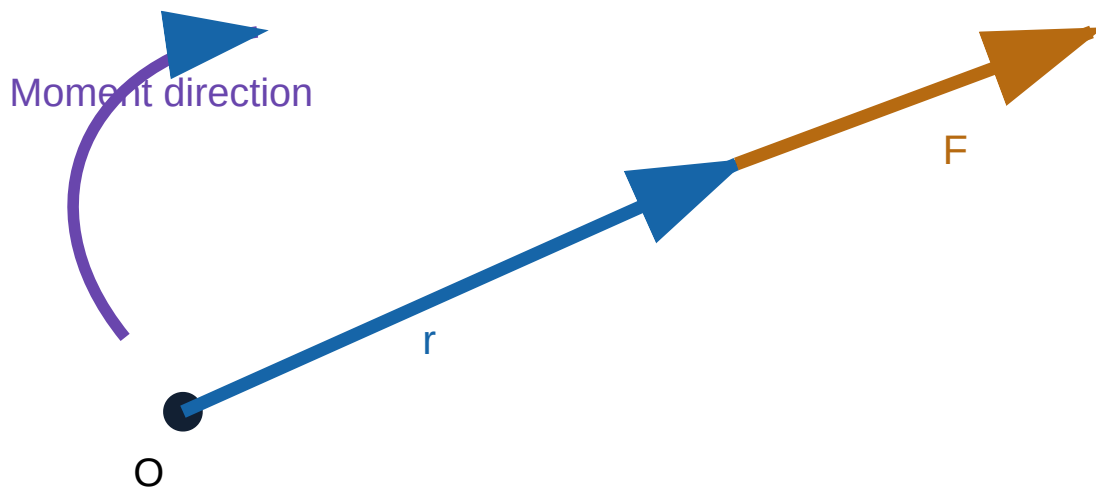
$$a \times b = 5i + 5j - 5k$$

$|a \times b| = 5\sqrt{3}$ , which equals the area of the parallelogram formed by  $a$  and  $b$ .

## 8. Moment of a force

$$\text{Moment about O: } M = r \times F$$

Here  $r$  is the position vector from the moment centre to any point on the line of action of  $F$ .



The moment vector is normal to the plane containing  $r$  and  $F$ . Its magnitude is force  $\times$  perpendicular distance.

## Solved Example 13: Moment

$r = 2i + j$  m and  $F = 3i + 4j$  N.

$$M = r \times F = (2 \times 4 - 1 \times 3)k = 5k \text{ N} \cdot \text{m}$$

**Answer:** 5 N·m in the positive  $k$  direction.

### Series Test I preparation

Practise one  $3 \times 3$  determinant, one matrix product, one Cramer problem, one inverse problem, one dot-product/work problem and one cross-product/moment problem.

1. Is speed scalar or vector? **Scalar.**
2. When are two nonzero vectors perpendicular? **When their dot product is zero.**
3. What does  $|a \times b|$  represent geometrically? **Area of the parallelogram.**
4. State the SI unit of moment. **N·m.**

MODULE 3 • CO3 • 18 HOURS

# Integral Calculus

Integration as inverse differentiation, standard results, substitution, integration by parts and definite integrals.

## 1. Integration as reverse differentiation

If  $dF/dx=f(x)$ , then F is an antiderivative of f and

$$\int f(x)dx = F(x)+C$$

The constant C is essential because the derivative of every constant is zero.

**Malayalam Support:** Differentiation-ന്റെ reverse operation ആണ് integration. Indefinite integral എഴുതുമ്പോൾ +C ഒഴിവാക്കരുത്.

### Verification

Differentiate your final answer. If its derivative equals the original integrand, the integration is correct.

## 2. Standard integrals

| Integrand                       | Integral                    | Condition / note  |
|---------------------------------|-----------------------------|-------------------|
| $x^n$                           | $x^{n+1}/(n+1)+C$           | $n \neq -1$       |
| $1/x$                           | $\ln x +C$                  | $x \neq 0$        |
| $e^x$                           | $e^x+C$                     | —                 |
| $a^x$                           | $a^x/\ln a + C$             | $a > 0, a \neq 1$ |
| $\sin x$                        | $-\cos x+C$                 | —                 |
| $\cos x$                        | $\sin x+C$                  | —                 |
| $\sec^2 x$                      | $\tan x+C$                  | —                 |
| $\operatorname{cosec}^2 x$      | $-\cot x+C$                 | —                 |
| $\sec x \tan x$                 | $\sec x+C$                  | —                 |
| $\operatorname{cosec} x \cot x$ | $-\operatorname{cosec} x+C$ | —                 |
| $1/(1+x^2)$                     | $\tan^{-1}x+C$              | —                 |
| $1/\sqrt{1-x^2}$                | $\sin^{-1}x+C$              | $ x  < 1$         |

### 3. Basic rules

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$$\int [af(x) \pm bg(x)] dx = a \int f(x) dx \pm b \int g(x) dx$$

#### Solved Example 14: Standard results

Evaluate  $\int (3x^2 - 4x + 5) dx$ .

$$= x^3 - 2x^2 + 5x + C$$

**Check:**  $d/dx(x^3 - 2x^2 + 5x + C) = 3x^2 - 4x + 5$ .

### 4. Integration by substitution

Substitution is useful when part of the integrand is the derivative of another part. Replace the inner expression by a new variable.

- 1 Choose  $u = g(x)$ .
- 2 Find  $du = g'(x) dx$ .
- 3 Rewrite the entire integral in  $u$ .
- 4 Integrate with respect to  $u$ .
- 5 Substitute back and add  $C$  for indefinite integrals.

#### Solved Example 15: Substitution

Evaluate  $\int 2x(x^2 + 1)^4 dx$ .

Let  $u = x^2 + 1$ ,  $du = 2x dx$ .

$$\int u^4 du = u^5/5 + C = (x^2 + 1)^5/5 + C$$

#### Solved Example 16: Logarithmic pattern

Evaluate  $\int (3x^2)/(x^3 + 5) dx$ .

Let  $u = x^3 + 5$ ,  $du = 3x^2 dx$ .

$$\int du/u = \ln|u| + C = \ln|x^3 + 5| + C$$

#### Common mistake

Do not substitute only the denominator. Every factor and  $dx$  must be rewritten consistently.

### 5. Integration by parts

$$\int u dv = uv - \int v du$$

Choose  $u$  using the practical priority ILATE: Inverse trigonometric, Logarithmic, Algebraic, Trigonometric, Exponential.



### Solved Example 17: $\int x e^x dx$

Choose  $u=x$ ,  $dv=e^x dx$ . Then  $du=dx$  and  $v=e^x$ .

$$\int x e^x dx = x e^x - \int e^x dx = e^x(x-1) + C$$

### Solved Example 18: $\int x \sin x dx$

$u=x$ ,  $dv=\sin x dx$ ;  $du=dx$ ,  $v=-\cos x$ .

$$\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C$$

## 6. Definite integrals

A definite integral has fixed lower and upper limits and gives a numerical value. If  $F'(x)=f(x)$ ,

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

**Malayalam Support:** Definite integral-ൽ +C വേണ്ട. Upper limit substitute ചെയ്ത value-ൽ നിന്ന് lower limit value കുറയ്ക്കണം.

### Limit reversal

$$\int_a^b f dx = -\int_b^a f dx$$

### Same limits

$$\int_a^a f dx = 0$$

### Interval splitting

$$\int_a^c f dx = \int_a^b f dx + \int_b^c f dx$$

### Linearity

Constants, sums and differences can be separated as in indefinite integration.

### Solved Example 19: Definite polynomial integral

Evaluate  $\int_0^2 (3x^2 + 2x) dx$ .

$$\left[ \frac{3}{4}x^2 \right]_0^2 = (8+4)-0=12$$

Answer: 12.

## Solved Example 20: Definite integral by substitution

Evaluate  $\int_0^1 2x(x^2+1) dx$ .

$u=x^2+1$ . When  $x=0$ ,  $u=1$ ; when  $x=1$ ,  $u=2$ .

$$\int_1^2 u du = \left[ \frac{u^2}{2} \right]_1^2 = (4-1)/2 = 3/2$$

## 7. Choosing a method

| Pattern                         | Preferred method                       | Example               |
|---------------------------------|----------------------------------------|-----------------------|
| Direct standard form            | Standard result                        | $\int x^5 dx$         |
| $g'(x)[g(x)]^n$ or $g'(x)/g(x)$ | Substitution                           | $\int 2x(x^2+1)^3 dx$ |
| Product of unlike functions     | By parts                               | $\int xe^x dx$        |
| Fixed limits                    | Find antiderivative, then apply limits | $\int_0^2 x^2 dx$     |

### Module 3 exam tip

Before integrating, simplify algebraically and identify the pattern. A correct method choice saves most of the work.

### Module 3 Quick Check and Answers

- $\int 4x^3 dx = x^4 + C$ .
- $\int \cos x dx = \sin x + C$ .
- Why is  $+C$  omitted in a definite integral? **It cancels in  $F(b)-F(a)$ .**
- State integration by parts.  **$\int u dv = uv - \int v du$ .**

## MODULE 4 • CO4 • 12 HOURS • SERIES TEST II

# Applications of Integration and Differential Equations

Area bounded by a curve and the axes; first-order, first-degree differential equations by variable separable and linear methods.

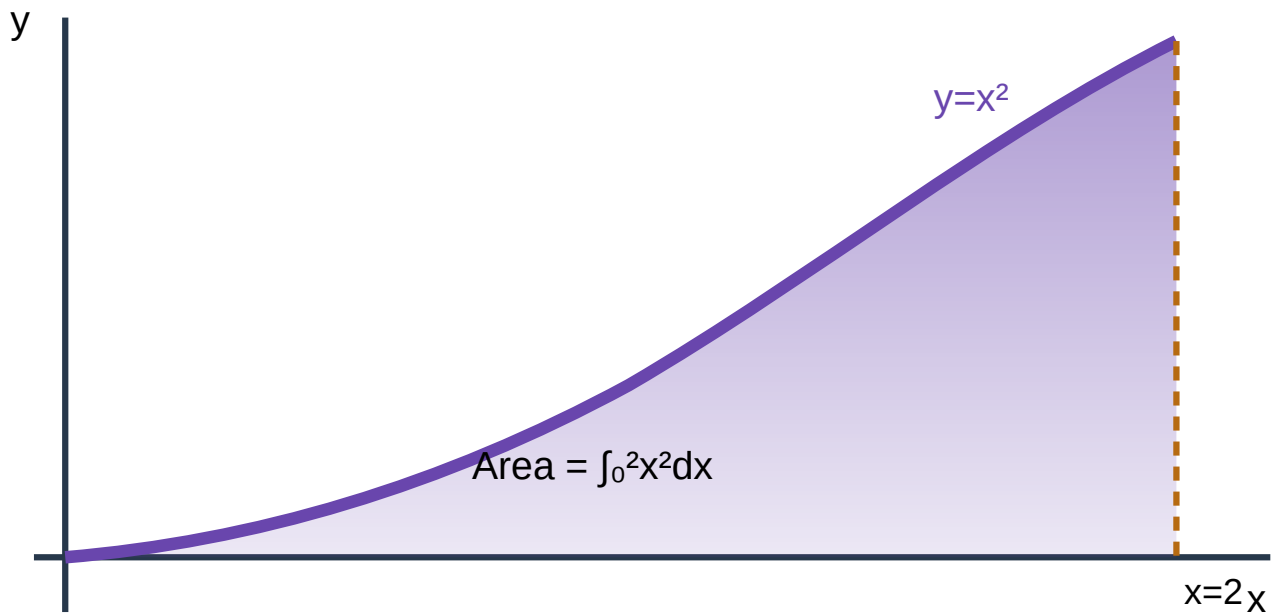
## 1. Area bounded by a curve and axes

If  $y=f(x)$  lies above the  $x$ -axis from  $x=a$  to  $x=b$ , the area is

$$A = \int_a^b y dx = \int_a^b f(x) dx$$

If the region is described more easily by  $x=g(y)$ , use  $A=\int x dy$ . Area is positive, so split the interval if the curve crosses an axis.

**Malayalam Support:** Curve- $\infty$  axis- $\infty$  ഇടയിലെ area കണ്ടെത്താൻ graph നോക്കി limits ശരിയായി എഴുതണം. Curve axis-ന്റെ താഴെയായിരിക്കിൽ integral negative ആകാം; geometric area positive ആയി എഴുതണം.



For  $y=x^2$ ,  $x=0$ ,  $x=2$  and the  $x$ -axis, the required area is the shaded region.

### Solved Example 21: Area under a parabola

Find the area bounded by  $y=x^2$ ,  $x=0$ ,  $x=2$  and the  $x$ -axis.

$$A = \int_0^2 x^2 dx = \left[ \frac{x^3}{3} \right]_0^2 = \frac{8}{3} \text{ square units}$$

### Solved Example 22: Area bounded by a line and axes

The line  $y=6-2x$  meets the  $y$ -axis at  $(0,6)$  and  $x$ -axis at  $(3,0)$ .

$$A = \int_0^3 (6-2x) dx = \left[ 6x - x^2 \right]_0^3 = 18 - 9 = 9 \text{ square units}$$

Check by triangle area:  $\frac{1}{2} \times 3 \times 6 = 9$ .

## 2. Differential equations: basic concepts

A differential equation contains derivatives of an unknown function. The **order** is the highest derivative order. The **degree** is the power of the highest-order derivative after the equation is polynomial in derivatives.

### First order

Highest derivative is  $dy/dx$ .

### First degree

The highest derivative occurs to power one.

### General solution

Contains an arbitrary constant.

**Particular solution**

Obtained after applying an initial/boundary condition.

**Solved Example 23: Order and degree**

$(dy/dx)^2 + y = x$  has order 1 and degree 2. The equation  $d^2y/dx^2 + 3dy/dx + y = 0$  has order 2 and degree 1.

**3. Variable separable equations**

An equation is separable when it can be rearranged so all y terms occur with dy and all x terms occur with dx.

$$dy/dx = f(x)g(y) \Rightarrow dy/g(y) = f(x)dx$$

1 Separate variables.

2 Integrate both sides.

3 Add one arbitrary constant.

4 Simplify the relation.

5 Use initial data if provided.

**Solved Example 24: Separable equation**

Solve  $dy/dx = xy$ .

$$dy/y = x dx$$

$$\ln|y| = x^2/2 + C \Rightarrow y = Ce^{x^2/2}$$

**Solved Example 25: Initial condition**

Solve  $dy/dx = 2xy$ ,  $y=3$  when  $x=0$ .

$$dy/y = 2x dx \Rightarrow \ln|y| = x^2 + C \Rightarrow y = Ce^{x^2}.$$

At  $x=0$ ,  $y=3$ , hence  $C=3$ .

**Answer:**  $y=3e^{x^2}$ .

**4. Linear differential equations**

The standard first-order linear form is

$$dy/dx + P(x)y = Q(x)$$

$$\text{Integrating factor (I.F.)} = e^{\int P(x)dx}$$

**Solution:**  $y \cdot \text{I.F.} = \int Q(x) \cdot \text{I.F.} \, dx + C$

**Malayalam Support:** Linear equation ആദ്യം  $dy/dx + Py = Q$  എന്ന standard form-ൽ എഴുതുക. ശേഷം  $\text{I.F.} = e^{\int P dx}$  കണ്ടെത്തി formula ഉപയോഗിക്കുക.

### Solved Example 26: Linear equation

Solve  $dy/dx + y = e^x$ .

$P=1$ ,  $Q=e^x$ .  $\text{I.F.} = e^{\int 1 dx} = e^x$ .

$$ye^x = \int e^x \cdot e^x dx + C = \int e^{2x} dx + C = e^{2x}/2 + C$$

$$y = e^x/2 + Ce^{-x}$$

### Solved Example 27: Linear equation with variable coefficient

Solve  $dy/dx + (1/x)y = x^2$ ,  $x > 0$ .

$\text{I.F.} = e^{\int (1/x) dx} = e^{\ln x} = x$ .

$$xy = \int x \cdot x^2 dx + C = x^4/4 + C$$

$$y = x^3/4 + C/x$$

## 5. Choosing the differential-equation method

| Form                               | Method               | First action                                  |
|------------------------------------|----------------------|-----------------------------------------------|
| $dy/dx = f(x)g(y)$                 | Variable separable   | Move y factors with dy and x factors with dx. |
| $dy/dx + P(x)y = Q(x)$             | Linear               | Identify P and Q; calculate I.F.              |
| Can be rearranged into either form | Choose shorter route | Simplify before deciding.                     |

### Series Test II preparation

Practise standard integrals, one substitution, one by-parts problem, one definite integral, one area problem, one separable equation and one linear equation.

### Module 4 Quick Check and Answers

1. What determines the order of a differential equation? **The highest derivative.**
2. State the I.F. for  $dy/dx + Py = Q$ .  **$e^{\int P dx}$ .**
3. Find area under  $y=x$  from 0 to 2. **2 square units.**
4. Is  $dy/dx = xy$  separable? **Yes.**

# Essential formulas with conditions

Source: 2002

## 2×2 determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

## 2×2 inverse

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Only if  $ad - bc \neq 0$ .

## Cramer's rule

$$x = \Delta x / \Delta, \quad y = \Delta y / \Delta, \quad z = \Delta z / \Delta$$

Requires  $\Delta \neq 0$ .

## Magnitude

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

## Dot product

$$a \cdot b = |a||b|\cos\theta = \sum_i a_i b_i$$

## Cross product

$$a \times b = |a||b|\sin\theta \hat{n}$$

## Work

$$W = F \cdot s$$

## Moment

$$M=r \times F$$

SI unit: N·m.

## Power rule

$$\int x^n dx = x^{n+1}/(n+1) + C$$

$n \neq -1$ .

## Log rule

$$\int dx/x = \ln|x| + C$$

## By parts

$$\int u \, dv = uv - \int v \, du$$

## Definite integral

$$\int_a^b f(x) dx = F(b) - F(a)$$

## Area

$$A = \int_a^b y \, dx$$

Split intervals if the curve crosses an axis.

## Separable DE

Copyright  
 $\frac{dy}{g(y)} = f(x)dx$

## Linear DE

$$dy/dx + Py = Q; IF = e^{\int P dx}$$

### METHOD BANK

## Problem-solving procedures

### 3×3 determinant

1. Choose a row or column.
2. Write + - + signs.
3. Form each 2×2 minor.
4. Simplify carefully.

### Matrix inverse solution

1. Write  $AX=B$ .
2. Find  $|A|$ .
3. Find  $A^{-1}$ .
4. Compute  $X=A^{-1}B$ .
5. Verify.

### Cramer's rule

1. Write  $\Delta$ .
2. Form  $\Delta_x, \Delta_y, \Delta_z$ .
3. Evaluate.
4. Divide by  $\Delta$ .
5. Check original equations.

### Vector application

1. Write components.
2. Choose dot or cross product.
3. Calculate.
4. State direction/unit.



## Integration

1. Simplify.
2. Identify standard/substitution/parts.
3. Integrate.
4. Add C or apply limits.
5. Differentiate to check.

## Differential equation

1. Identify separable or linear form.
2. Apply method.
3. Integrate.
4. Add C.
5. Use condition and verify.

### IMPORTANT DIAGRAM AND GRAPH BANK

## Visual ideas to reproduce in answers

| Visual                 | Must show                                                     | Why it matters                        |
|------------------------|---------------------------------------------------------------|---------------------------------------|
| Matrix multiplication  | Row of first matrix, column of second, product entry          | Prevents order mistakes               |
| Parallelogram law      | a, b and resultant a+b                                        | Shows vector addition geometrically   |
| Dot product angle      | Two vectors and included angle $\theta$                       | Connects product with projection/work |
| Cross product / moment | r, F and perpendicular moment direction                       | Shows right-hand rule and torque      |
| Area under curve       | Curve, axes, limits and shaded region                         | Helps choose correct integral         |
| Linear DE flow         | Standard form $\rightarrow$ IF $\rightarrow$ solution formula | Organises steps                       |

### SOLVED PROBLEM BANK

## Additional examination-style worked examples

### S1. Solve two equations by inverse method

$$2x+3y=13, \quad x+2y=8.$$

$$A=[2 \ 3; 1 \ 2], \quad |A|=1, \quad A^{-1}=[2 \ -3; -1 \ 2].$$

$$X=A^{-1}B=[2 \ -3; -1 \ 2][13; 8]=[2; 3]$$

**Answer:**  $x=2, y=3$ .

## S2. Cramer's rule

Solve  $x+y+z=6$ ,  $x-y+z=2$ ,  $2x+y-z=1$ .

$\Delta=6$ ,  $\Delta x=6$ ,  $\Delta y=12$ ,  $\Delta z=18$ .

**Answer:**  $x=1$ ,  $y=2$ ,  $z=3$ .

## S3. Work by a force

$F=5i+2j-k$  N,  $s=3i-j+4k$  m.

$$W=F \cdot s=15-2-4=9 \text{ J}$$

## S4. Moment

$r=i+2j+3k$  m,  $F=2i-j+k$  N.

$$M=r \times F=5i+5j-5k \text{ N} \cdot \text{m}$$

## S5. Substitution

$\int x/\sqrt{x^2+4} \, dx$ . Let  $u=x^2+4$ ,  $du=2x \, dx$ .

$$= \frac{1}{2} \int u^{-1/2} du = \sqrt{u} + C = \sqrt{x^2+4} + C$$

## S6. By parts

$\int x \ln x \, dx$ . Let  $u=\ln x$ ,  $dv=x \, dx$ .

$$= x^2 \ln x / 2 - \frac{1}{2} \int x \, dx = x^2 \ln x / 2 - x^2 / 4 + C$$

## S7. Definite integral

$\int_0^{\pi/2} \sin x \, dx = [-\cos x]_0^{\pi/2} = 1$ .

## S8. Area

Area under  $y=4-x^2$  from  $x=0$  to  $x=2$ :

$$A = \int_0^2 (4-x^2) dx = [4x - x^3/3]_0^2 = 16/3 \text{ square units}$$

## S9. Separable DE

$dy/dx = x/y$ .

$$y \, dy = x \, dx \Rightarrow y^2/2 = x^2/2 + C \Rightarrow y^2 = x^2 + C$$

## S10.1 Linear DE

$$dy/dx+2y=4.$$

$$IF=e^{2x}.$$

$$ye^{2x}=\int 4e^{2x}dx+C=2e^{2x}+C \Rightarrow y=2+Ce^{-2x}$$

### EXPECTED QUESTIONS

## Module-wise high-priority practice

These are fresh syllabus-based practice patterns, not a claim about the exact examination paper.

### Module 1

- Define determinant, matrix and order.
- Evaluate second- and third-order determinants.
- List and identify matrix types.
- Add, subtract and multiply matrices.
- Solve three equations by Cramer's rule.
- Solve two equations using inverse matrix.

### Module 2

- Differentiate scalar and vector quantities.
- Find magnitude and unit vector.
- Find vector between two points.
- Use dot product to find angle or work.
- Use cross product to find area or moment.
- Interpret product sign and direction.

### Module 3

- Explain integration as reverse differentiation.
- State standard integral results.
- Solve polynomial and trigonometric integrals.
- Apply substitution.
- Apply integration by parts.
- Evaluate definite integrals.

### Module 4

- Set limits from a graph.
- Find area bounded by curve and axes.
- Define order and degree.
- Solve separable equations.

- Solve linear first-order equations.

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- Apply an initial condition.

## PRACTICE SECTION

# Objective, short-answer and problem practice

## A. Objective questions

1. The determinant of  $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$  is: (a) 5 (b) 8 (c) 11 (d) -5.
2. A matrix with all diagonal entries 1 and all others 0 is: (a) zero (b) identity (c) row (d) rectangular.
3. If  $\mathbf{a} \cdot \mathbf{b} = 0$  for nonzero vectors, they are: (a) parallel (b) equal (c) perpendicular (d) opposite.
4.  $|\mathbf{a} \times \mathbf{b}|$  gives the area of a: (a) triangle (b) circle (c) parallelogram (d) rectangle only.
5.  $\int dx/x$  equals: (a)  $1/x^2$  (b)  $\ln|x| + C$  (c)  $x^2/2$  (d)  $e^x$ .
6. The I.F. of  $dy/dx + 3y = x$  is: (a)  $e^x$  (b)  $e^{3x}$  (c)  $3e^x$  (d)  $x^3$ .

## B. Short problems

7. Evaluate  $\begin{vmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 1 & 1 \end{vmatrix}$ .
8. Find  $AB$  for  $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & 1 \\ 4 & -1 \end{bmatrix}$ .
9. Find a unit vector along  $2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ .
10. Find work when  $\mathbf{F} = 2\mathbf{i} + 3\mathbf{j}$  N and  $\mathbf{s} = 4\mathbf{i} - \mathbf{j}$  m.
11. Evaluate  $\int (4x^3 + 2/x) dx$ .
12. Evaluate  $\int_0^1 (x^2 + 1) dx$ .
13. Find area under  $y = 2x$  from  $x = 0$  to  $x = 3$ .
14. Solve  $dy/dx = 3x^2y$ .
15. Solve  $dy/dx + y = 1$ .

## C. Long/application problems

16. Solve  $x + y + z = 6$ ,  $2x + y - z = 1$ ,  $x - y + 2z = 5$  by Cramer's rule.
17. Solve  $4x + y = 11$ ,  $2x + 3y = 13$  using inverse matrix.
18. For  $\mathbf{r} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$  and  $\mathbf{F} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ , find moment and magnitude.
19. Evaluate  $\int x^2 e^x dx$  using repeated integration by parts.
20. Find the area bounded by  $y = x(4 - x)$ , the  $x$ -axis,  $x = 0$  and  $x = 4$ .

## Practice final answers

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1. 5
2. Identity
3. Perpendicular
4. Parallelogram
5.  $\ln|x|+C$
6.  $e^{3x}$
7. -1
8.  $[10 \ -1; 12 \ -3]$
9.  $(2i-2j+k)/3$
10. 5 J
11.  $x^4+2\ln|x|+C$
12.  $4/3$
13. 9 square units
14.  $y=Ce^{x^3}$
15.  $y=1+Ce^{-x}$
16.  $x=1, y=2, z=3$
17.  $x=2, y=3$
18.  $M=-i+5j+7k; |M|=\sqrt{75}$
19.  $e^x(x^2-2x+2)+C$
20.  $32/3$  square units

## Self-assessment

- ✓ I can identify the order and type of a matrix.
- ✓ I can evaluate a  $3 \times 3$  determinant without sign errors.
- ✓ I can decide between dot and cross product.
- ✓ I can choose standard, substitution or by-parts integration.
- ✓ I can read integration limits from a graph.
- ✓ I can distinguish separable and linear differential equations.

### QUICK REVISION

## Last-day revision sheet

### Determinant

$2 \times 2$ :  $ad-bc$ ;  $3 \times 3$ : + - + expansion.

### Inverse

Exists only if determinant is nonzero.

### Matrix product

Inner dimensions must match.

### Dot product

Scalar result; angle and work.

### Cross product

Vector result; area and moment.

### Integration

Reverse differentiation; include +C.

### Definite integral

$F(b)-F(a)$ , no +C in final value.

### Area

Sketch, select limits, integrate positive region.

### Separable DE

Separate x and y before integrating.

### Linear DE

Standard form, I.F., then solution formula.

### Ten common mistakes

Wrong cofactor sign; changing variable order in Cramer's rule; multiplying matrices entry-by-entry; forgetting  $|A| \neq 0$ ; confusing dot and cross products; omitting units for work/moment; forgetting  $+C$ ; incorrect substitution differential; applying limits in reverse; using the linear-DE formula before standardising the equation.

### FRESH SYLLABUS-BASED MODEL

# Model Question Paper

The paper follows the common 75-mark diploma theory structure and is newly written from the official Mathematics II syllabus.

## MATHEMATICS II - 2002

Time: 3 Hours | Maximum Marks: 75

**Instructions:** Answer as directed. Show essential working. Assume suitable missing data only where necessary. Use exact values unless approximation is requested.

### PART A

5×1=5

Answer all questions.

1. Evaluate  $\begin{vmatrix} 3 & 1 & 2 \\ 4 & & \end{vmatrix}$ .
2. State the condition for the inverse of a square matrix to exist.
3. What is the dot product of perpendicular nonzero vectors?
4. Write  $\int \cos x \, dx$ .
5. State the integrating factor of  $dy/dx + P(x)y = Q(x)$ .

### PART B

5×8=40

Answer any five questions.

6. (a) Evaluate  $\begin{vmatrix} 2 & 1 & 3 & 1 & 0 & 2 & 3 & 2 & 1 \end{vmatrix}$ . (b) Add and multiply  $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & & \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & 0 & 1 \\ 4 & & \end{bmatrix}$ .
7. Solve  $x+y+z=6$ ,  $2x-y+z=3$ ,  $x+2y-z=2$  by Cramer's rule.
8. For  $a=2i-j+2k$  and  $b=i+3j-k$ , find  $a \cdot b$ ,  $a \times b$  and the angle between them.
9. A force  $F=3i+4j-2k$  N acts at position  $r=2i-j+k$  m. Find the moment about the origin and its magnitude.
10. Evaluate (a)  $\int 2x(x^2+4)^3 dx$  and (b)  $\int x e^x dx$ .
11. Evaluate (a)  $\int_0^2 (x^2+2x) dx$  and (b) find the area under  $y=3x$  from  $x=0$  to  $x=2$ .
12. Solve (a)  $dy/dx=xy$  and (b)  $dy/dx+2y=6$ .

### PART C

2×15=30

Answer any two questions.

13. (a) Explain important matrix types with examples. (b) Solve  $3x+y=7$  and  $2x+y=5$  by inverse method. (c) Verify the answer.
14. (a) Explain scalar and vector products. (b) A force  $F=4i-2j+3k$  N moves a point through  $s=2i+j-4k$  m; find work. (c) At  $r=i+2j$  m, a force  $3i+4j$  N acts; find moment.
15. (a) Evaluate  $\int x \sin x \, dx$ . (b) Find the area bounded by  $y=4-x^2$ , the x-axis and  $x=0$  in the first quadrant. (c) Solve  $dy/dx + (1/x)y = x^2$ ,  $x>0$ .

### COMPLETE MODEL ANSWERS

## Part A

1.  $3 \times 4 - 1 \times 2 = 10$ .

2. The determinant must be nonzero:  $|A| \neq 0$ .

3. Zero.

4.  $\sin x + C$ .

5.  $e^{\int P(x) dx}$ .

## Part B

### 6. Determinant and matrix operations

(a)  $\Delta = 2(0 \times 1 - 2 \times 2) - 1(1 \times 1 - 2 \times 3) + 3(1 \times 2 - 0 \times 3) = -8 + 5 + 6 = 3$ .

(b)  $A+B = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$ .

$$AB = \begin{bmatrix} 1 \times 2 + 2 \times 1, & 1 \times 0 + 2 \times 4; & 3 \times 2 + 1 \times 1, & 3 \times 0 + 1 \times 4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 7 & 4 \end{bmatrix}$$

### 7. Cramer's rule

$\Delta = 7$ ,  $\Delta x = 7$ ,  $\Delta y = 14$ ,  $\Delta z = 21$ .

**Answer:**  $x=1$ ,  $y=2$ ,  $z=3$ . Substitution satisfies all three equations.

### 8. Vector products and angle

$a \cdot b = 2 \times 1 + (-1) \times 3 + 2 \times (-1) = -3$ .

$$a \times b = -5i + 4j + 7k$$

$|a|=3$ ,  $|b|=\sqrt{11}$ .  $\cos\theta = -3/(3\sqrt{11}) = -1/\sqrt{11}$ .

**Answer:**  $\theta = \cos^{-1}(-1/\sqrt{11})$ .

### 9. Moment

$$M = r \times F = \begin{bmatrix} i & j & k \\ 2 & -1 & 1 \\ 3 & 4 & -2 \end{bmatrix} = -2i + 7j + 11k$$

$|M| = \sqrt{4 + 49 + 121} = \sqrt{174} \text{ N}\cdot\text{m}$ .

## 10. Integration

(a)  $u=x^2+4$ ,  $du=2x \, dx$ .

$$\int u^3 du = u^4/4 + C = (x^2+4)^4/4 + C$$

(b)  $u=x$ ,  $dv=e^x dx$ .

$$\int x e^x dx = x e^x - e^x + C = e^x(x-1) + C$$

## 11. Definite integral and area

(a)  $[x^3/3 + x^2]_0^2 = 8/3 + 4 = 20/3$ .

(b)  $A = \int_0^2 3x \, dx = [3x^2/2]_0^2 = 6$  square units.

## 12. Differential equations

(a)  $dy/y = x \, dx \Rightarrow \ln|y| = x^2/2 + C \Rightarrow y = Ce^{x^2/2}$ .

(b)  $I F = e^{2x} \cdot y e^{2x} = \int 6e^{2x} dx + C = 3e^{2x} + C$ . Hence  $y = 3 + Ce^{-2x}$ .

## Part C

### 13. Matrices and inverse method

(a) A row matrix has one row; column matrix one column; rectangular matrix unequal rows and columns; square matrix equal rows and columns; zero matrix all entries zero; diagonal matrix non-diagonal entries zero; scalar matrix equal diagonal entries; identity matrix diagonal entries one and other entries zero.

(b)  $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ ,  $X = \begin{bmatrix} x \\ y \end{bmatrix}$ ,  $B = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$ .  $|A| = 1$ .

$$A^{-1} = \begin{bmatrix} 1 & -1 & -2 \\ 3 & 2 & 1 \end{bmatrix}, X = A^{-1}B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(c) **Verification:**  $3(2)+1=7$  and  $2(2)+1=5$ . Therefore  $x=2, y=1$ .

### 14. Scalar/vector products, work and moment

(a) Dot product  $a \cdot b = |a||b|\cos\theta$  is a scalar and is used for angle, projection and work. Cross product  $a \times b = |a||b|\sin\theta \hat{n}$  is a vector perpendicular to both and is used for area and moment.

(b)  $W = F \cdot s = 4 \times 2 + (-2) \times 1 + 3 \times (-4) = -6 \, \text{J}$ .

(c)  $r = i + 2j$ ,  $F = 3i + 4j$ .

$$M = r \times F = (1 \times 4 - 2 \times 3)k = -2k \, \text{N} \cdot \text{m}$$

The negative  $k$  direction indicates clockwise sense in the  $xy$ -plane.

### 15. Integration, area and linear differential equation

(a)  $u=x, dv=\sin x \, dx$ . Then  $v=-\cos x$ .

$$\int x \sin x \, dx = -x \cos x + \sin x + C$$



(b) In the first quadrant,  $y=4-x^2$  meets x-axis at  $x=2$ .

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$$A = \int_0^2 (4-x^2) dx = [4x - x^3/3]_0^2 = 16/3 \text{ square units}$$

(c)  $P=1/x$ , IF  $=e^{\int dx/x} = x$ .

$$xy = \int x \cdot x^2 dx + C = x^4/4 + C \Rightarrow y = x^3/4 + C/x$$

## ANSWER KEY

# Compact answer index

| Area         | Key answers / checks                                                                                     |
|--------------|----------------------------------------------------------------------------------------------------------|
| Module 1     | Determinant sign pattern + - +; AB exists when inner dimensions match; inverse exists for $ A  \neq 0$ . |
| Module 2     | $a \cdot b$ scalar; $a \times b$ vector; work $= F \cdot s$ ; moment $= r \times F$ .                    |
| Module 3     | Always add C to indefinite integrals; definite integral uses upper minus lower.                          |
| Module 4     | Area needs graph and limits; separable equations separate variables; linear equations use IF.            |
| Practice MCQ | 1-a, 2-b, 3-c, 4-c, 5-b, 6-b.                                                                            |

## OFFICIAL REFERENCES

# Textbooks and online resource guidance

| Code | Reference                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------|
| T1   | B. S. Grewal, <i>Higher Engineering Mathematics</i> , Khanna Publishers, New Delhi, 40th Edition, 2007.    |
| R1   | G. B. Thomas and R. L. Finney, <i>Calculus and Analytic Geometry</i> , Addison Wesley, 9th Edition, 1995.  |
| R2   | Reena Garg, <i>Engineering Mathematics</i> , Khanna Publishing House, New Delhi, Revised Edition, 2018.    |
| R3   | S. S. Sabharwal and Sunita Jain, <i>Applied Mathematics, Vol. I &amp; II</i> , Eagle Parkashan, Jalandhar. |

### Online resources listed by the syllabus

Use e-books, e-tools, learning mathematics software and suitable mathematics websites for additional visualisation and practice. The official syllabus does not specify individual web addresses.

## FINAL EXAM REVISION CHECKLIST

# Before entering the examination hall

## Concept checklist

- ✓ Determinant order and  $3 \times 3$  sign pattern
- ✓ Matrix types and multiplication condition
- ✓ Dot versus cross product
- ✓ Standard integrals and method selection
- ✓ Area limits from graph
- ✓ Separable versus linear DE

## Presentation checklist

- ✓ Write formula before substitution
- ✓ Keep variable order consistent
- ✓ Show vector units and direction
- ✓ Add  $+C$  where required
- ✓ Apply upper minus lower limit
- ✓ Box the final answer and verify

### Final reminder

Mathematics marks are earned through correct method and visible working. Even when the final arithmetic is wrong, a clearly stated formula and correct intermediate steps may receive partial credit.

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